Question Paper

Exam Date & Time: 20-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CSS 2101/CSS_2101 DATA STRUCTURES
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) An AVL Tree is created by inserting the following keys in Chronological order: (5)
O, G, T, E, L, Y, Z.
After successful insertion, the key G is deleted.
Analyze and show:
i. The structure of the AVL tree before and after deletion.
ii. Identify the type of imbalance.Specify the rotation(s) used to restore balance.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

1B) Suppose two queues, Q1 and Q2, are implemented in a single array of size 10 using the method where Q1 grows from left to right and Q2 grows from right to left. Initially, both are empty. If 4 elements are enqueued into Q1 and 3 elements into Q2, what are the possible indices of front1 and front2? Demonstrate your answer with a neat diagram. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

1C) Write a C program to define a structure Student containing another structure Date (for date of birth). Read and display the student details using nested structures. The structure should resemble the following table: (2)

ID	Student Name	DoB		
		DD	MM	YYYY
1	Ram	01	01	2020
2	Lakshman	01	01	2022
3	Hanuma	01	01	2025

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2A) A college is developing a simple "**Smart Editor**" software for its students. (5)
In the **Smart Editor**, students should be able to**undo and redo** text changes.As a software developer, you are asked to analyze and illustrate how **Stacks** can be used in this application.
Question:
i. Explain how a **Stack** can be used to implement the *Undo-Redo* feature in the editor.Write short C code snippets for implementation and explain how it will work.

Course Outcome: "CO3" / Cognitive Level: "Application"

2B) Consider the following example to create binary search tree: (3)
40, 20, 10, 30, 60, 50, 70.
Illustrate with the above BST, how a null pointer is replaced to convert it into a double threaded binary tree.

Course Outcome: "CO4" / Cognitive Level: "Application"

2C) Assume that you are storing marks of students in 3 subjects. (2)
Write a C program that:
i. Takes the number of students as input.
ii. Dynamically allocates a **2D array** using pointers.
iii. Accepts marks for each subject.Calculates and prints the**average marks** of each student using only **pointer arithmetic.**

Course Outcome: "CO1" / Cognitive Level: "Application"

3A) i. Write a function to perform the Fast Transpose computation of a given sparse matrix. (5)
ii. Demonstrate complete step-by-step process to compute the transpose of the below given sparse matrix given in Fig.Q.3A using the above function. Also show the contents of all the variables **(row_terms and starting_pos)** used during the computation process.

0	0	3	0	4	0
0	0	5	7	0	0
0	0	0	0	0	9
0	2	6	0	0	0
1	0	0	0	0	8
0	0	4	0	0	0

Fig.Q.3A

Course Outcome: "CO1" / Cognitive Level: "Application"

3B) In a gaming leaderboard system, players' scores are organized in a Binary Search Tree (BST) to enable quick ranking and retrieval. Write an iterative function to insert a new player's score into the BST. (3)
Insert the score 65 into the following BST and illustrate the updated tree structure after insertion.

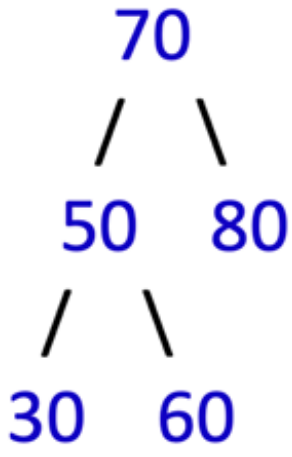


Fig. Q. 3B

Course Outcome: "CO4" / Cognitive Level: "Application"

3C) With the help of the following code, justify why a stack is preferred over a queue in handling nested function calls during program execution. Also, explain the role of activation records (stack frames) in this process (2)
void fun1() { printf("Inside fun1n"); }

```
void fun2() { fun1(); printf("Inside fun2n"); }
int main() { fun2(); printf("Inside mainn"); }
```

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4A) A music streaming service keeps track of songs in a playlist using a singly linked list. Each node stores the following details: SongID (integer), Title (string), Duration (in seconds). (5)

Write a C program to perform the following operations:

Insert songs into the playlist in the order they are added.

Remove all songs whose duration is less than a user-specified value (Use head pointer for performing this operation).

Display the updated playlist.

Include a main() function to demonstrate all functionalities.

Note:- Use the following prototypes:-

```
typedef struct node {
   int songID;
   char title[50];
   int duration;
   struct node *next;
} Node;
Node* insertSong(Node *head, int songID, char title[], int duration);
Node* removeShortSongs(Node *head, int minDuration);
void displayPlaylist(Node *head);
```

Course Outcome: "CO2" / Cognitive Level: "Application"

4B) Build the expression tree for the given infix expression 8+4*2/(3+1)-5 . Clearly show all the intermediate steps (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

4C) You are provided with the array representation of a binary tree. (2)

Array Representation: [1, 4, 2, 10, 5, 9, 13] Array index starts with 1.

Using the given representation, apply your understanding of tree structures to answer the following:

i. Construct the binary tree from the given array and determine the depth of node 9.

ii. State the formula to find the index of right child given the index of parent and use it to find the index of right child of node 2.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5A) A small delivery network is represented as an undirected graph. The network has six hubs labelled A, B, C, D, E, F. The direct connections between hubs are: A-B, A-C, B-D, C-E, D-F, E-F, B-C, D-E (5)

Note: Whenever there is a choice between multiple unvisited neighbours in DFS or BFS, hubs should be selected in alphabetical order.

i. If a delivery inspection follows Depth First Search (DFS) starting from hub A, which hub will be inspected last? Explain your reasoning.

ii. If a delivery inspection follows Breadth First Search (BFS) starting from hub A, which hub is the farthest from A in terms of the number of connections?

iii. Determine the number of connected subnetworks in this network.

Now, remove the link E-F and D-F from the network. How many connected subnetworks exist after these removals?

iv. If all connections become one-way (directed) in the order: A→B→D→F, A→C→E→F, B→C, D→E, can DFS starting from A reach all hubs? Provide a brief explanation.

v. Which change in the underlying data structure (queue ↔ stack) would make BFS behave like

DFS?

Course Outcome: "CO5" / Cognitive Level: "Analysis"

5B) Develop a C program using a Doubly linked list to manage a queue system in a hospital, where patients can register, be served in order, or leave the queue. (3)

```
struct Patient {
int id;
char name [50];
struct Patient *next; };
Function prototypes:
void registerPatient(int id, char name[])
void servePatient() //as per queue order
void removePatient(int id)
```

Course Outcome: "CO2" / Cognitive Level: "Application"

5C) Write a C function void rearrangeQueue(int queue[], int *front, int *rear) that rearranges the queue such that all even numbers appear before all odd numbers while preserving their original relative order within their respective groups (evens and odds). (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

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Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2101:DATA STRUCTURES

#	Course Outcome	Description	Marks
1	1	CO 1	9
2	2	CO 2	8
3	3	CO 3	12
4	4	CO 4	16
5	5	CO 5	5
		Total	**50**

* NA - CO Not Selected

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END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2101:DATA STRUCTURES

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	9	32(64%)
4	Analysis	6	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)